

Harsh Shah

Boston, MA | harshbshah9@gmail.com | 617-943-0430 | [linkedin.com/in/shahharsh11](https://www.linkedin.com/in/shahharsh11)

Summary

Software Engineer with 6+ years of experience seeking mid-level Software Engineer and Full Stack Engineer roles. Proficient across the full stack Angular, React, TypeScript on the frontend; Node.js, Java, C#/.NET and Python on the backend; and AWS and Azure for cloud infrastructure and event-driven architectures. Track record delivering production enterprise platforms across financial services, B2B e-commerce, IoT domains — with recent experience building RAG pipelines and agentic workflows.

Technical Skills

Web & Mobile: React, React Native, Angular, Android, Next.js, Magento, Vue.js, HTML, CSS, SCSS, Tailwind CSS, Expo, Vuetify, Leaflet

Languages: JavaScript, TypeScript, Java, C#, Kotlin, Python, C++, Go

Backend: Node.js, C#/.NET, GraphQL, Java, Spring Boot, PHP, Express, Hapi.js, REST,

Cloud/DevOps: AWS(Lambda, API Gateway, Cognito, S3, RDS, Bedrock), Azure(Functions, Event Hub, AD B2C, Service Bus, IoT Hub), Docker, Kubernetes, GitHub actions

Databases: PostgreSQL, MySQL, SQLite, SQL Server, MongoDB, Redis, DynamoDB, Pinecone, Qdrant, OpenSearch

AI/ML: RAG, LangChain, LangSmith, pgvector, Prompt Engineering, FastAPI, OpenAI API

Professional Experience

Valtech | Detroit, MI, USA

June 2021 - Current

Software Engineer

Connected Fire Safety Dashboard

- Built a role-based real time data analysis **Angular** dashboard monitoring alarms, device health and sensor telemetry across **1,000+** LoRa RF-enabled gateways generating **millions** of daily sensor data points, visualized time-series data with Highcharts and exported real time data to XLSX reports
- Implemented multi-tenant hierarchy across districts, customers, sites, systems and nodes with **Azure MSAL** authentication via **AD B2C**, **i18n** localization and Application Insights for frontend monitoring
- Built the **C#/.NET** Azure Functions backend pipeline consisting of an **Event Hub** processor ingests real-time telemetry into a **Service Bus** queue, a data processor function handles fault detection, and a health monitoring timer runs every 15 minutes to detect missed heartbeats and gateway disconnections, triggering push, email, and SMS alerts via **Azure Notification Hubs**

B2B HVAC E-Commerce Platform

- Developing a large scale **Magento** B2B e-commerce platform serving **2000+** HVAC contractors and distributors, implemented custom modules, search autocomplete with product suggestions powered by **Azure AI search** and theme customizations
- Reduced ordering time for HVAC distributors by implementing a Quick Order workflow supporting bulk CSV uploads with Magento and custom checkout flow with **Paymetric payment gateway** integration
- Integrated SX API (ERP **PHP** backend) for real-time pricing, inventory, order history and customer credit data, built custom pricing display and SKU availability modules consuming live ERP data

Loan and County Tax Management Platform

- Built and maintained a **React** reporting platform using **AG Grid** Enterprise with server-side row model, custom cell editors, and 20+ configurable report types; developed a custom **jsPDF** reporting engine with grouping, sorting, advanced cell rendering and XLSX export capabilities

Enterprise CMS Modernization

- Modernized Fidelity's **Sitecore** based web platform using **Angular**, **Node.js** and **EJS** templates improving website performance and page load times by **40%**

Smart HVAC Mobile App

- Owned development of a production **React Native** + **Expo** app with **10,000+** active users; built geofencing automation, multi-thermostat management and push alerts and integrated **GraphQL** + **Apollo Client** backed by **AWS** serverless

V2X Safety Android Library & App

- Kotlin Android** library + companion app for SAE J2735 BLE-based V2X safety messaging; **AWS Cognito auth**, **S3** storage, **API Gateway** endpoints, and **Vue.js** map dashboard for message analysis

Ascent FutureTech | Mumbai, MH, India

July 2017 - July 2018

Associate Software Engineer

- Built a full stack product management dashboard using **Angular** and **Spring Boot**, designed **REST** APIs end-to-end covering order creation, status tracking and delivery workflows and integrated with a relational backend for order history and inventory state
- Contributed to a freelancer marketplace using **React** (frontend) and **Python/Flask** (REST APIs) implemented service listing, bidding, and contract workflows; integrated **Braintree payment gateway** with **webhook** handling for payment lifecycle events, contributing to a **16%** increase in successful transactions

Education

Northeastern University | *Boston, MA, USA*

September 2018 - December 2020

Master of Science in Information Systems

Mumbai University | *Mumbai, MH, India*

July 2013 - June 2017

Bachelor of Engineering in Information Technology

Independent Projects

Document Q&A Agent (Python, FastAPI, LangChain, pgvector, Claude API, RAGAS)

May 2026

- Built an end-to-end **RAG** pipeline ingesting PDFs via **PyMuPDF** chunking with recursive splitting, embedding via OpenAI text-embedding-3-small into **pgvector** (Supabase), with hybrid BM25 + vector retrieval fused via RRF which improved context recall from 0.64 → 0.78 over pure vector search
- Implemented agentic tool calling using the Anthropic **Claude API** for multi-step reasoning with **LangSmith** tracing, enforced hallucination grounding via XML structured prompts and citation requirements achieving faithfulness score of 0.84 on a 70 item eval dataset
- Built **RAGAS** regression suite measuring faithfulness, answer relevancy, context precision and used eval results to tune chunking strategy and retrieval weights, exposed via **FastAPI**

Bike Rental Application (MEAN Stack)

December 2020

- Built a location based bicycle rental web app using **Angular, Node.js, and MongoDB** with JWT authentication and secure password hashing, integrated **Google Maps API** for nearby station search/navigation and **Spotify API** for in-app music playback

Opinion Mining for Hotel Reviews (Android)

May 2017

- Built an **Android** app that analyzes hotel reviews and converts them into ratings using **PoS tagging** and **TF-IDF based sentiment analysis**, enabled hotel search, review submission and automated rating generation based on keyword scoring